

DISTRIBUTED MULTINOMIAL REGRESSION

- $j = 1, 2, \dots, d$: the category index
- $i = 1, 2, \dots, n$: the observation index
- c_{ij} : the i -th observation in j -th category
- v_i : p -dimensional attribute vector on observation unit i of n total
- $m_i = \sum_j c_{ij}$: the total count for observation i across categories
- $\eta_{ij} = \alpha_j + v_i^T \phi_j$: the generalized linear model
- $e^{\eta_{ij}}$: the Poisson indensity drawn for each observation c_{ij}
- $\Lambda_i = \sum_{k=1}^d e^{\eta_{ik}}$
- $q_{ij} = \frac{e^{\eta_{ij}}}{\Lambda_i}$: the probability of category j for observation i

We connect attributes v_i and counts through a big multinomial logistic regression model

$$p(c_i | v_i; m_i) = \text{MN}(c_i; q_i, m_i).$$

Normal multinomial regression model costs more when the category count d is large. For each observation c_{ij} we need to normalize the probability p_{ij} by calculating $\frac{e^{\eta_{ij}}}{\sum_{k=1}^d e^{\eta_{ik}}}$, thus leads to the expensive cost. The log-likelihood of the multinomial model is

$$\sum_{i=1} \log(m_i!) - \sum_{i=1} \sum_{j=1} \log(c_{ij}!) + \sum_{i=1} \sum_{j=1} c_{ij} \log(e^{\eta_{ij}}) - \sum_{i=1} m_i \log(\Lambda_i).$$

The methodology of simplifying the multinomial regression model is replacing Λ_i by our “new” estimator, then do d individual Poisson surrogate models.

STRATEGY

The simple multinomial $\text{MN}(m_i, q_i)$, where q_{ij} denotes the probability for c_{ij} . The log-likelihood of the model that up to an additive constant is (dropped the constant terms $\log(m_i!) - \sum_j \log(c_{ij}!)$)

$$\sum_i \sum_j c_{ij} \log(q_{ij})$$

and the softmax is $\log(q_{ij}) = \eta_{ij} - \log(\Lambda_i)$.

Hence, the log-likelihood is

$$\sum_i \sum_j c_{ij} [\eta_{ij} - \log(\Lambda_i)] = \sum_i \sum_j [c_{ij} \eta_{ij} - c_{ij} \log(\Lambda_i)].$$

For the first term, we simplify the calculation by matrix computation, where:

$$\sum_i \sum_j c_{ij} \eta_{ij} = \sum_i c_i^T \eta_i.$$

For the second term, Λ_i is independent on j and $m_i = \sum_j c_{ij}$, thus,

$$\begin{aligned} \sum_i \sum_j c_{ij} \log(\Lambda_i) &= \sum_i \log(\Lambda_i) \sum_j c_{ij} \\ &= \sum_i m_i \log(\Lambda_i). \end{aligned}$$

Hence, the negative log-likelihood of the single multinomial model is

$$\sum_i [m_i \log(\Lambda_i) - c_i^T \eta_i].$$

We replace η_{ij} by $\eta_{ij} + \mu_i$, where μ_i denotes the observation fixed effect and it only corresponds with observation label i , not with group label j . The new negative log-likelihood of new “shift” model is

$$\begin{aligned} \sum_i [m_i \log(\sum_j e^{\eta_{ij} + \mu_i}) - c_i^T (\eta_i + \mu_i)] &= \sum_i [m_i (\mu_i + \log(\sum_j e^{\eta_{ij}})) - c_i^T (\eta_i + \mu_i)] \\ &= \sum_i [m_i \mu_i + m_i \log(\Lambda_i) - c_i^T \eta_i - \mu_i \sum_j c_{ij}] \\ &= \sum_i [m_i \log(\Lambda_i) - c_i^T \eta_i]. \end{aligned}$$

which implies that the shift does not change the log-likelihood. In statistical model, we say that the softmax cares about the differences across categories, but not the shift totally. This property that not identifiable has the following meaning,

1. Non-uniqueness of parameters: Some models that without constraint could express the same log-likelihood and then the same statistical model, which suggests the parameter redundancy because one degree of freedom of “shift” has been wrongly kept.
2. Need identifiability constraints: Because of non-uniqueness, the model must be anchored by an additional constraint, such as choosing a baseline category or imposing a sum-to-zero condition. Without such a constraint, the model is not identifiable.
3. Parameters are interpretable only relatively.
4. Consequences for optimisation: Lead to singular Hessians, unstable optimisation, slow convergence, or non-unique estimates.
5. Comparison structure: The fitted probabilities depend only on how categories compare with one another, not on any common shift applied to all of them.

The Poisson model unconditional on m_i is defined as follows,

$$c_{ij} \sim P(\lambda_{ij}),$$

where $\log(\lambda_{ij}) = \eta_{ij} + \mu_i$. Since the log-likelihood of $Y \sim P(\lambda)$ is $Y \log(\lambda) - \lambda - \log(Y!)$, the negative log-likelihood (remove constant term) for all c_{ij} is the sum of i and j ,

$$\begin{aligned} \sum_i \sum_j \lambda_{ij} - c_{ij} \log(\lambda_{ij}) &= \sum_i \sum_j e^{\eta_{ij} + \mu_i} - c_{ij} (\eta_{ij} + \mu_i) \\ &= \sum_i e^{\mu_i} \sum_j e^{\eta_{ij}} - \sum_i \sum_j c_{ij} \eta_{ij} - \sum_i \mu_i \sum_j c_{ij} \\ &= \sum_i e^{\mu_i} \Lambda_i - \sum_i c_i^T \eta_i - \sum_i m_i \mu_i \end{aligned}$$

Let

$$g(\mu_i) = \sum_i e^{\mu_i} \Lambda_i - \sum_i c_i^T \eta_i - \sum_i m_i \mu_i,$$

the gradient $\frac{\partial}{\partial \mu_i} g(\mu_i) = e^{\mu_i} \Lambda_i - m_i$ for fixed i , then the MLE for μ_i in Poisson model is estimated at its conditional MLE,

$$\mu_i^* = \log(m_i) - \log(\Lambda_i).$$

However, if we want to surrogate the multinomial model by Poisson models, μ_i^* creates computational bottleneck, because the term $\log(\Lambda_i) = \log(\sum_j e^{\eta_{ij}})$ for each category j need to consider all η_{ij} and be updated across categories. Hence, we suggest a plug-in estimator $\hat{\mu}_i = \log(m_i)$ and the corresponding $g(\hat{\mu}_i) = m_i(\sum_j e^{\eta_{ij}} - 1)$. Define the plug-in MLEs $\hat{\eta}_i = [\hat{\eta}_{i1}, \dots, \hat{\eta}_{id}]^T$ which minimize the Poisson negative log-likelihood function under $\mu = \hat{\mu}_i$, then $m \hat{\mu}_i$ is the optimal joint MLE estimate for i , and thus that $\hat{i}, \hat{i}$ minimize the Poisson objective while $\hat{i}$ minimizes the logistic multinomial objective.